TAMIL NADU STATE BOARD - STANDARD 10 SCIENCE

CHAPTER 3: BIOLOGY

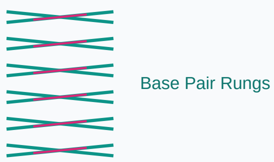
PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 10 English Medium
Subject:	Science
Chapter Name:	Biology

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

Double Helix DNA Segment Structure



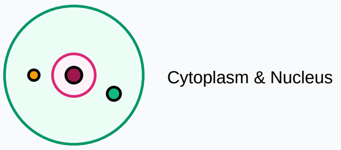
Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Photosynthesis	The process by which green plants utilize sunlight, carbon dioxide, and water to synthesize glucose and release oxygen.
Double Circulation	A circulation system where blood flows through the heart twice in one complete cycle of body circulation.
Hormone	Chemical messengers secreted directly into the bloodstream by endocrine glands to regulate physiological functions.
Heredity	The transmission of genetic characters from parents to offspring through DNA replication and chromosomal inheritances.

Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

Eukaryotic Cell Organelle Layout



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:

Biology System Representation

Essential Formulas, Laws & Identities:

- Photosynthesis reaction: $6CO_2 + 6H_2O + \text{Light} \rightarrow C_6H_{12}O_6 + 6O_2$
- Cardiac Output: Cardiac Output = Stroke Volume * Heart Rate
- Mendel Monohybrid Phenotypic Ratio: 3 : 1 | Genotypic Ratio: 1 : 2 : 1

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: If a person's stroke volume is 70 ml and heart rate is 72 beats per minute, calculate their cardiac output.

Step-by-step Solution:
Apply formula: Cardiac Output = Stroke Volume * Heart Rate.
Cardiac Output = 70 * 72 = 5040 ml per minute, which is approx 5.04 liters/min.

Biology System Representation

HOTS (Higher Order Thinking Skills) Challenge

Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: Explain why pea plants were chosen by Gregor Mendel for his genetic hybridization experiments.

Analytical Solution Strategy:
Mendel selected garden peas (Pisum sativum) due to: (1) short life cycle, (2) presence of clear contrasting characters, (3) easy cross-pollination artificially, (4) hermaphrodite flowers facilitating natural self-pollination.

Biology System Representation

Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

Q1 (1-Mark): Describe the structure of human heart with a labelled schematic representation.

Q2 (2-Mark): What is a reflex arc? List its components.

Q3 (2-Mark): Explain the functions of the plant hormone Auxin.

Q4 (5-Mark): Describe the structure of DNA double helix as proposed by Watson and Crick.

Q5 (5-Mark): What are the main causes and preventive measures for Diabetes Mellitus?

Biology System Representation

Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Agricultural Hormone Treatment

Auxin and Gibberellin sprays are utilized in commercial orchards to trigger seedless fruit development (parthenocarpy).

Application Case: Genetic Diagnostics

Understanding DNA structure allows labs to sequence chromosomes and screen for hereditary conditions.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

Biology System Representation

Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.

✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.

✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3, N) for the final numeric answer.

✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

[ ] Read and memorized core definitions on Page 2.

[ ] Reviewed critical formulas and relations on Page 3.

[ ] Solved worked examples on Page 4.

[ ] Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress. Good luck!

Biology System Representation